3 MILLION TECHNICAL TALENT (3MTT) NIGERIA

DATA SCIENCE CAPSTONE PROJECT REPORT

WAEC PASS LIKELIHOOD PREDICTION USING MACHINE LEARNING

A Demonstration Using a Student Mathematics Performance Dataset

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DECLARATION

Declaration

I declare that this capstone project titled “WAEC Pass Likelihood Prediction Using Machine Learning” is my original work completed as part of the requirements for the 3 Million Technical Talent (3MTT) Data Science Programme.

Where information, libraries, datasets, or published materials have been used, appropriate acknowledgement has been made. This work has not been submitted elsewhere for any other academic or professional certification.

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ABSTRACT

Educational institutions often face challenges in identifying students who are at risk of failing examinations before final results are released. Early prediction of student performance enables timely academic intervention, helping educators provide additional support where needed.

This project, titled "WAEC Pass Likelihood Prediction Using Machine Learning," present s a machine learning approach for predicting whether a student is likely to pass or fail an examination. Since official WAEC candidate datasets are not publicly available for research purposes, this study uses the publicly available Student Mathematics Performance Dataset as a case study to demonstrate the complete data science workflow.

The dataset was cleaned, explored, and pre-processed before training three machine learning classification models: Logistic Regression, Decision Tree, and Random Forest. Model performance was evaluated using accuracy, confusion matrices, and classification reports.

Among the evaluated models, Logistic Regression achieved the highest accuracy of 94.94%, outperforming Random Forest (91.14%) and Decision Tree (88.61%). The findings demonstrate that machine learning techniques can effectively predict student examination outcomes and identify learners who may require additional academic support.

Although this study uses a Mathematics performance dataset rather than official WAEC records, the methodology presented can be applied to authentic WAEC datasets whenever such data becomes available.

Keywords: Machine Learning, Data Science, WAEC, Student Performance, Classification, Predictive Analytics, Educational Data Mining.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Education plays a vital role in the social and economic development of every nation. Students' academic performance is one of the key indicators used to evaluate the effectiveness of teaching, learning, and educational policies. In Nigeria, the West African Seni or School Certificate Examination (WAEC) is one of the most important examinations used to assess students' readiness for higher education and employment opportunities. The outcome of this examination significantly influences students' academic and career paths.

Despite the importance of WAEC, many students perform below expectations due to factors such as poor study habits, high absenteeism, previous academic difficulties, limited parental support, and socio-economic challenges. Unfortunately, these challenges are often identified only after examination results have been released, reducing opportunities for timely intervention.

The emergence of Data Science and Machine Learning has created new opportunities to analyse educational data and predict students' academic outcomes before final examinations. By identifying patterns within historical student data, machine learning models can estimate the likelihood of students passing or failing, enabling educators to provide targeted support to those at risk.

This project applies machine learning techniques to predict students' likelihood of passing examinations. Since official WAEC datasets are not publicly available for research purposes, this study uses a publicly available Student Mathematics Performance dataset as a case study. Although the dataset focuses on Mathematics, it contains educational and behavioural variables that effectively demonstrate the complete machine learning workflow required for predicting examination outcomes.

The project follows a standard data science process involving data collection, pre-processing, exploratory data analysis, model development, evaluation, and comparison of multiple machine learning algorithms to determine the most suitable model for predicting student performance.

1.2 Problem Statement

Every year, a significant number of students perform poorly in external examinations such as WAEC. Many schools rely on periodic assessments and teacher observations to identify struggling students, but these methods may not detect students who require academic support early enough.

The lack of predictive tools limits the ability of educators to make informed decisions before examinations take place. Without early identification, schools may miss opportunities to implement interventions that could improve students' academic outcomes.

This project addresses this challenge by developing a machine learning model capable of predicting the likelihood of students passing or failing based on relevant educational factors. Such predictions can support data-driven decision-making and early intervention strategies.

1.3 Aim of the Study

The aim of this project is to develop a machine learning model capable of predicting students' likelihood of passing examinations using educational data.

1.4 Objectives of Objectives

The objectives of this project are to:

1.Collect and analyse a student performance dataset.

2.Pre-process and prepare the dataset for machine learning.

3.Perform exploratory data analysis to identify important patterns and relationships.

4.Develop machine learning classification models for predicting student pass likelihood.

5.Compare the performance of Logistic Regression, Decision Tree, and Random Forest algorithms.

6.Identify the model with the highest predictive performance.

7.Provide recommendations based on the findings.

1.5 Research Questions

This study seeks to answer the following questions:

1.Can machine learning accurately predict students' likelihood of passing examinations?

2.Which student-related factors have the greatest influence on academic performance?

3.Which machine learning algorithm performs best for this prediction task?

4.How can predictive analytics support educational decision-making?

1.6 Scope of the Study

This project focuses on predicting students' likelihood of passing examinations using machine learning techniques. The study applies data preprocessing, exploratory data analysis, model development, and performance evaluation on a publicly available Student Mathematics Performance dataset.

Note on Dataset Selection: The capstone project title, "WAEC Pass Likelihood Prediction Using Machine Learning," was provided as part of the programme requirements. However, official WAEC candidate datasets are not publicly available for public research due to privacy and accessibility limitations. Therefore, this study uses a Student Mathematics Performance dataset as a case study to demonstrate the required data science workflow. The methodology developed in this project can be applied to official WAEC datasets when such data becomes available.

1.7 Significance of the Study

This project is significant because it demonstrates how machine learning can be applied to educational data to support early identification of students who may require additional academic assistance.

The findings may benefit:

1.Students, by identifying those who may need early intervention.

2.Teachers, by providing insights into factors associated with academic performance.

3.School administrators, by supporting data-driven educational planning.

4.Researchers, by serving as a reference for future studies in educational data mining.

5.Policy makers, by highlighting the potential of artificial intelligence in improving educational outcomes.

1.8 Limitations of the Study

One limitation of this project is the use of a publicly available Student Mathematics Performance dataset instead of official WAEC examination records. Consequently, the developed model demonstrates the machine learning methodology using Mathematics per formance data rather than actual WAEC candidate data.

Additionally, the dataset contains a limited number of student records and variables, which may not capture all factors influencing examination performance. Future studies can improve the model by incorporating larger datasets and official WAEC records, subject to availability and ethical approval.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to student performance prediction, educational data mining, and the application of machine learning in education. It also discu sses the machine learning algorithms used in this project and highlights previous studi es relevant to predicting academic performance.

2.2 Educational Data Mining

Educational Data Mining (EDM) is a field that applies data science, statistics, and machi ne learning techniques to educational data in order to understand student learning patt erns and improve educational outcomes.

Educational institutions generate large volumes of data, including attendance records, examination scores, assignment grades, demographic information, and behavioural re cords. Analysing these data enables educators to identify students who may be at risk of poor academic performance and implement timely interventions.

Educational Data Mining has become an important area of research because it support s evidence-based decision-making and helps improve teaching effectiveness.

2.3 Machine Learning in Education

Machine Learning is a branch of Artificial Intelligence (AI) that enables computer syste ms to learn patterns from historical data and make predictions without being explicitly programmed for every task.

In education, machine learning is applied to:

1.Predict student academic performance.

2.Identify students at risk of failure.

3.Recommend personalised learning resources.

4.Detect student dropout risks.

5.Improve institutional planning and resource allocation.

Predictive models help schools move from reactive decision-making to proactive interv ention by identifying students who may require additional support before final examina tions.

2.4 Classification in Machine Learning

Classification is a supervised machine learning technique used to predict categorical o utcomes. It learns from labelled historical data and classifies new observations into pr edefined categories.

In this project, classification is used to predict whether a student is likely to:

Pass (1)

Fail (0)

The target variable, Pass, was created from the final Mathematics grade (G3), where:

Pass = 1 if the final grade is 10 or above

Fail = 0 if the final grade is below 10

This transformed the prediction problem into a binary classification task.

2.5 Machine Learning Algorithms Used

2.5.1 Logistic Regression

Logistic Regression is a supervised classification algorithm used for predicting binary outcomes. It estimates the probability that an observation belongs to a particular class.

In this project, Logistic Regression predicts the probability that a student will pass or fai l based on academic and demographic features.

Advantages include:

Easy to interpret.

Fast training time.

Suitable for binary classification.

Performs well on structured datasets.

The results of this study showed that Logistic Regression achieved the highest predicti on accuracy among the evaluated models.

2.5.2 Decision Tree

Decision Tree is a supervised learning algorithm that predicts outcomes by splitting dat a into branches based on feature values.

The model resembles a tree structure consisting of:

Root node

Decision nodes

Leaf nodes

Decision Trees are easy to understand and visualise but may suffer from overfitting when not properly controlled.

2.5.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting.

Each tree is trained on a random sample of the dataset, and the final prediction is determined through majority voting.

Random Forest generally provides:

Higher accuracy

Better generalisation

Improved robustness

Reduced overfitting

2.6 Related Studies

Several researchers have applied machine learning techniques to predict student academic performance.

Cortez and Silva (2008) developed predictive models using secondary school student performance data and demonstrated that academic and social factors significantly influence students' final grades.

Romero and Ventura (2013) highlighted the growing importance of Educational Data Mining in improving teaching strategies and supporting educational decision-making through predictive analytics.

Research has also shown that factors such as previous academic performance, study habits, attendance, parental education, and learning environment contribute significantly to student success.

These studies demonstrate that machine learning is an effective tool for analysing educational data and predicting academic outcomes.

2.7 Research Gap

Although several studies have explored student performance prediction using machine learning, limited publicly available research focuses specifically on WAEC examination prediction because official examination datasets are not readily accessible.

This project addresses that limitation by demonstrating a complete machine learning workflow using a publicly available Student Mathematics Performance dataset. The approach can be adapted for official WAEC datasets when such data becomes available.

2.8 Summary of Literature Review

The literature reviewed indicates that machine learning has become an effective approach for predicting student academic performance. Educational Data Mining enables institutions to analyse educational data and identify students who may require early intervention.

Among the algorithms commonly used for classification, Logistic Regression, Decision Tree, and Random Forest are widely recognised for their effectiveness. These algorithms form the foundation of the predictive models developed and evaluated in this project.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted in developing the machine learning model for predicting students' likelihood of passing examinations. It covers the dataset used, data collection, pre-processing, exploratory data analysis, model development, ev aluation methods, and the software tools used during the project.

3.2 Research Methodology

This study adopted a quantitative and experimental research approach using supervise d machine learning classification techniques.

The methodology followed the standard Data Science lifecycle, which includes:

1.Data Collection

2.Data Understanding

3.Data Pre-processing

4.Exploratory Data Analysis (EDA)

5.Feature Engineering

6.Model Development

7.Model Evaluation

8.Model Comparison

9.Interpretation of Results

3.3 Dataset Description

The dataset used in this project is the Student Performance Mathematics Dataset, whic h is publicly available through the UCI Machine Learning Repository.

Although the project title focuses on WAEC pass likelihood prediction, an official WAEC dataset was not publicly available for research. Therefore, this dataset was selected b ecause it contains educational variables that closely relate to student academic perfor mance.

The dataset consists of:

395 student records

33 variables (features)

Some of the important variables include:

Variable Description

Sex	Student gender
Age	Student age
Study time	Weekly study time
Failures	Number of previous failures
Schools up	Extra educational support
Famsup	Family educational support
Internet	internet access at home
Absences	Number of school absences
G1	First period Mathematics grade
G2	Second period Mathematics grade
G3	Final Mathematics grade

3.4 Software and Tools Used

The project was implemented using the following tools:

Tool	Purpose
GoogleColab	Writing and executing Python code
Python	Programming language
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Matplotlib	Data visualisation
Scikit-learn	Machine learning model development
GitHubVersion	control and project documentation

3.5 Data Pre-processing

Before building the machine learning models, the dataset was cleaned and prepared.

The pre-processing steps included:

Step 1: Importing the Dataset

The Student Mathematics Performance dataset was imported into Google Colab using the Pandas library.

Step 2: Data Inspection

The dataset was examined using:

Head()

Shape

Info()

Describe()

These functions helped identify the structure of the dataset and confirm that there were 395 rows and 33 columns, with no missing values.

Step 3: Creating the Target Variable

A new target variable called Pass was created from the final Mathematics grade (G3).

The classification rule used was:

Pass = 1 if G3 ≥ 10

Fail = 0 if G3 < 10

This transformed the problem into a binary classification task.

Step 4: Encoding Categorical Variables

Machine learning algorithms require numerical input. Therefore, categorical variables such as gender, school, family support, and internet access were converted into numerical values using Label Encoding.

Step 5: Splitting the Dataset

The processed dataset was divided into:

80% Training Data

20% Testing Data

The training data was used to train the machine learning models, while the testing data was used to evaluate their performance.

3.6 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand patterns within the dataset before model development.

The following analyses were performed:

Distribution of students who passed and failed

Relationship between study time and pass rate

Relationship between previous failures and pass rate

Distribution of student absences

Correlation analysis of numerical variables

The visualisations provided insights into the factors associated with student performance.

3.7 Model Development

Three supervised machine learning classification algorithms were developed and trained:

Logistic Regression

Logistic Regression was used as the baseline classification model because it is suitable for binary prediction tasks.

Decision Tree

A Decision Tree classifier was trained to learn decision rules from the educational data.

Random Forest

Random Forest combined multiple decision trees to improve predictive performance and reduce overfitting.

Each model was trained using the same training dataset to ensure a fair comparison.

3.8 Model Evaluation

The performance of each machine learning model was evaluated using the following metrics:

Accuracy

Accuracy measures the percentage of correct predictions made by the model.

Confusion Matrix

The confusion matrix was used to compare predicted results with actual outcomes and identify correctly and incorrectly classified students.

Classification Report

The classification report included:

Precision

Recall

F1-score

Support

These metrics provided a more detailed assessment of model performance.

3.9 Model Comparison

The three developed models were compared based on their prediction accuracy.

The comparison showed that:

Model	Accuracy
Logistic Regression	94.94%
Random Forest	91.14%
Decision Tree	88.61%

Based on these results, Logistic Regression was selected as the best-performing model.

3.10 Summary

This chapter presented the methodology adopted in this study. It described the dataset, pre-processing procedures, exploratory data analysis, machine learning models, evaluation metrics, and model comparison. The next chapter presents the results obtained from the developed models and discusses their implications.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from the implementation of the machine learning models developed in this study. It includes the outcomes of the exploratory data analysis (EDA), model evaluation, comparison of machine learning algorithms, and discussion of the findings. The objective is to assess the effectiveness of different machine learning techniques in predicting students' likelihood of passing examinations.

4.2 Dataset Overview

The Student Mathematics Performance dataset was successfully loaded into Google Colab and examined before model development.

The dataset contained:

Item	Value
Number of Students	395
Number of Features	33
Missing Values	0
Target Variable	Pass (Created from G3)

The absence of missing values meant that no imputation was required before training the models.

4.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the characteristics of the dataset and identify factors associated with student performance.

4.3.1 Pass and Fail Distribution

The target variable (Pass) was created from the final Mathematics grade (G3).

Students with a final grade of 10 or above were classified as Pass (1), while students w ith a grade below 10 were classified as Fail (0).

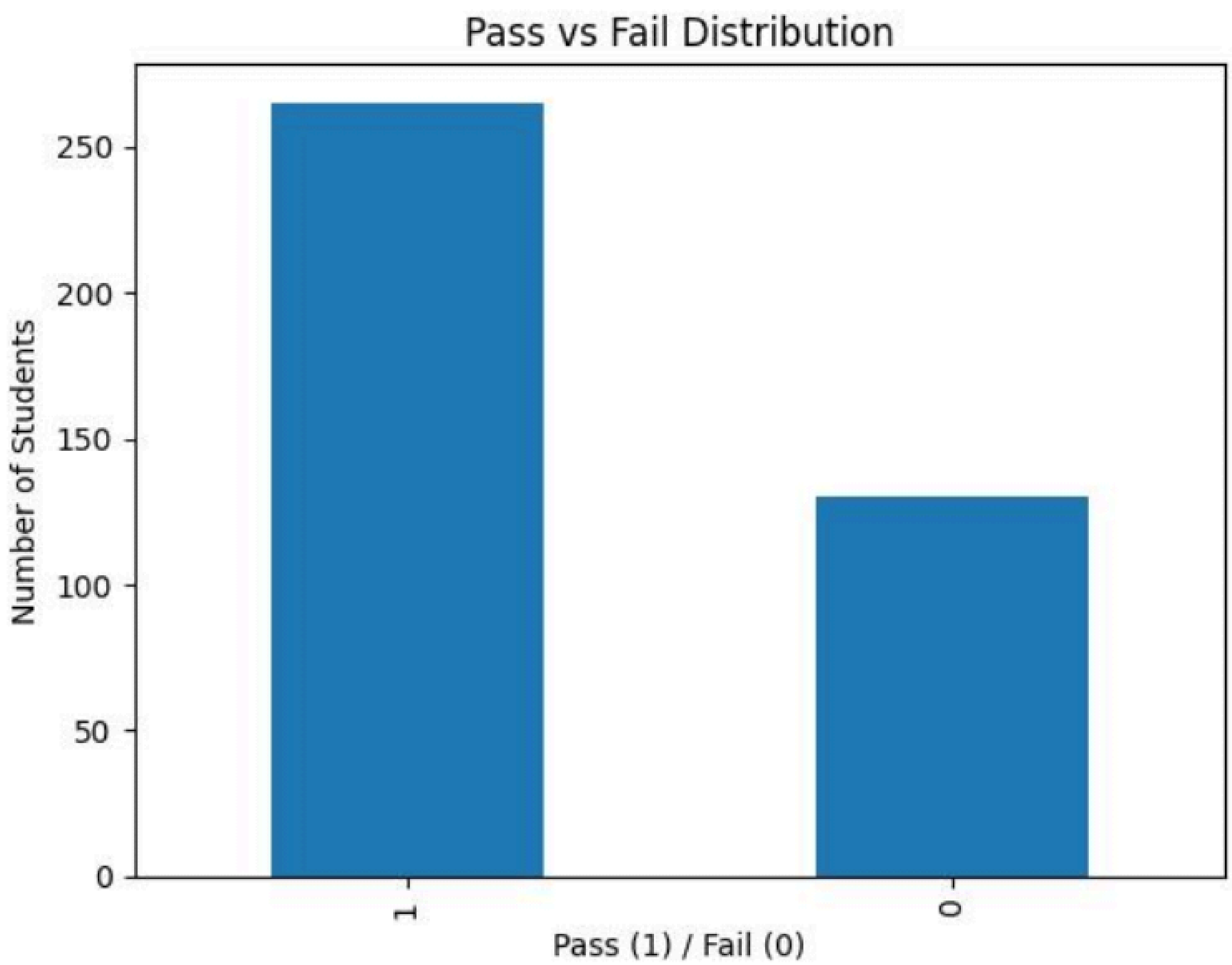
The analysis showed:

Outcome	Number of Students
Pass	265
Fail	130

Interpretation

The dataset contains more students who passed than failed, indicating a moderately balanced dataset that is suitable for binary classification.

Figure 4.1: Pass vs Fail Distribution



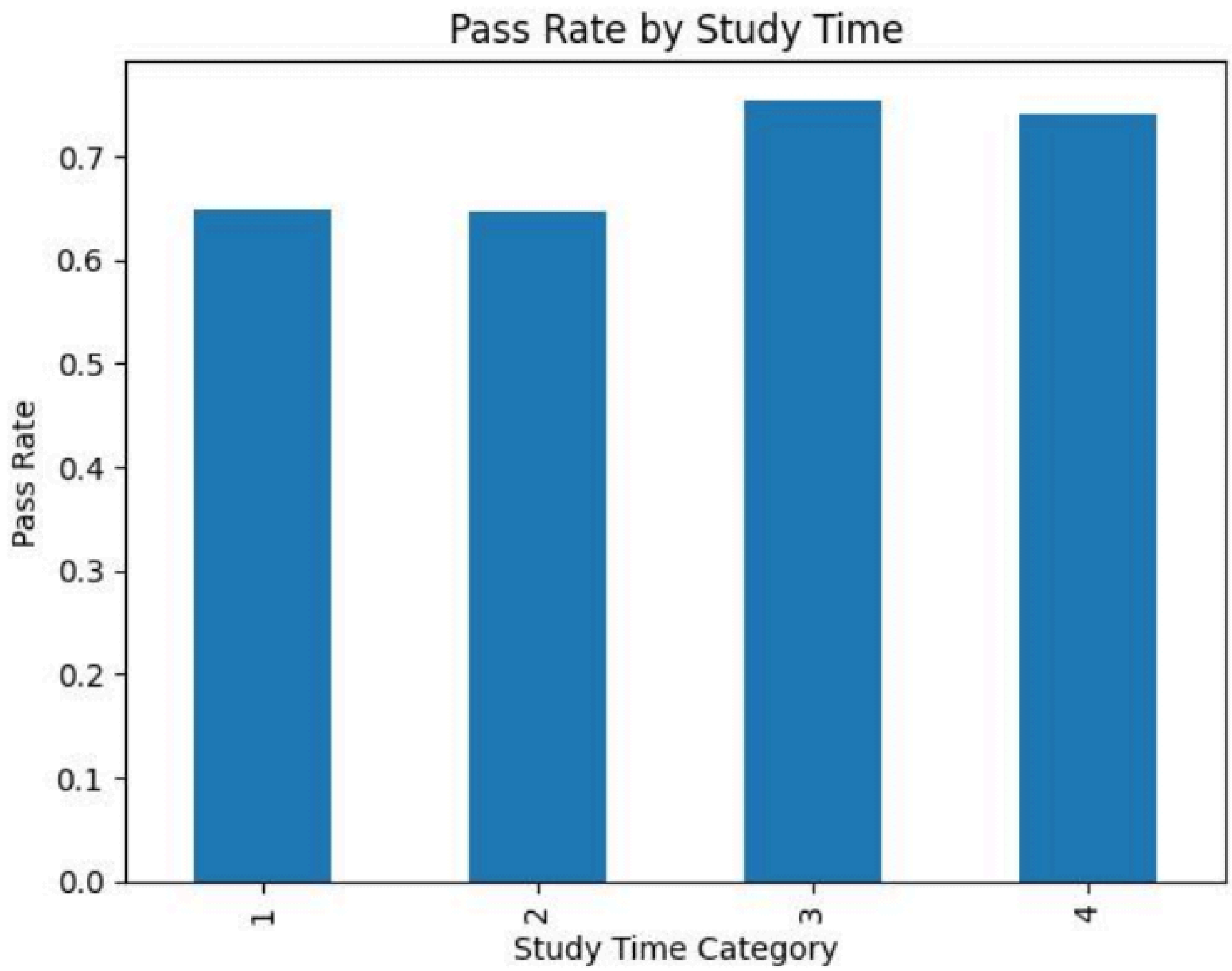
4.3.2 Study Time and Pass Rate

The relationship between study time and pass rate was analysed.

The results indicated that students who reported spending more time studying generall y achieved higher pass rates than those who studied less.

This suggests that study habits play an important role in academic performance.

Figure 4.2: Study Time vs Pass Rate



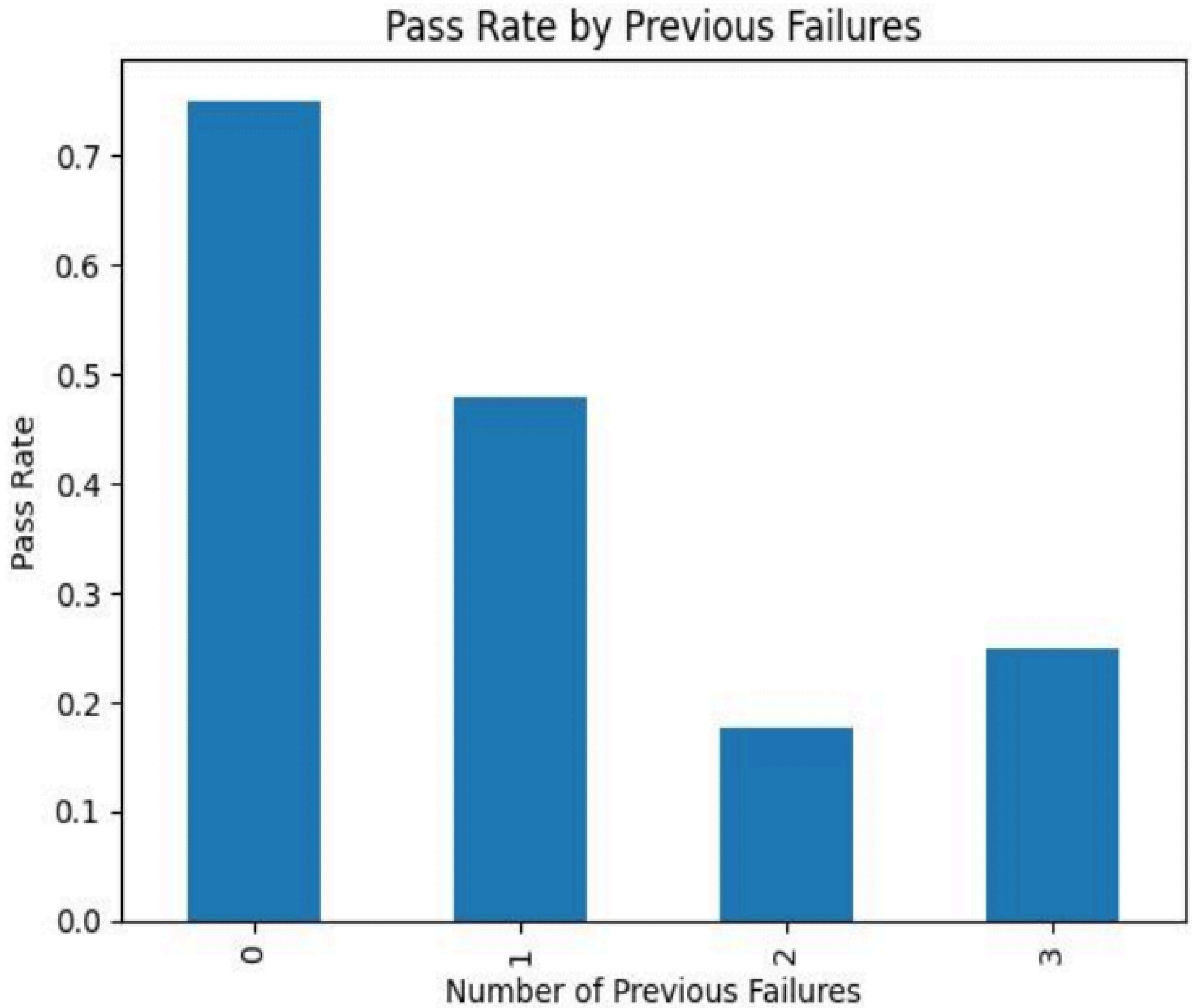
4.3.3 Previous Failures and Pass Rate

The analysis examined how previous academic failures influenced students' likelihood of passing.

Students with no previous failures had the highest pass rate, while students with multip le previous failures were more likely to fail.

This indicates that previous academic performance is an important predictor of future success.

Figure 4.3: Previous Failures vs Pass Rate

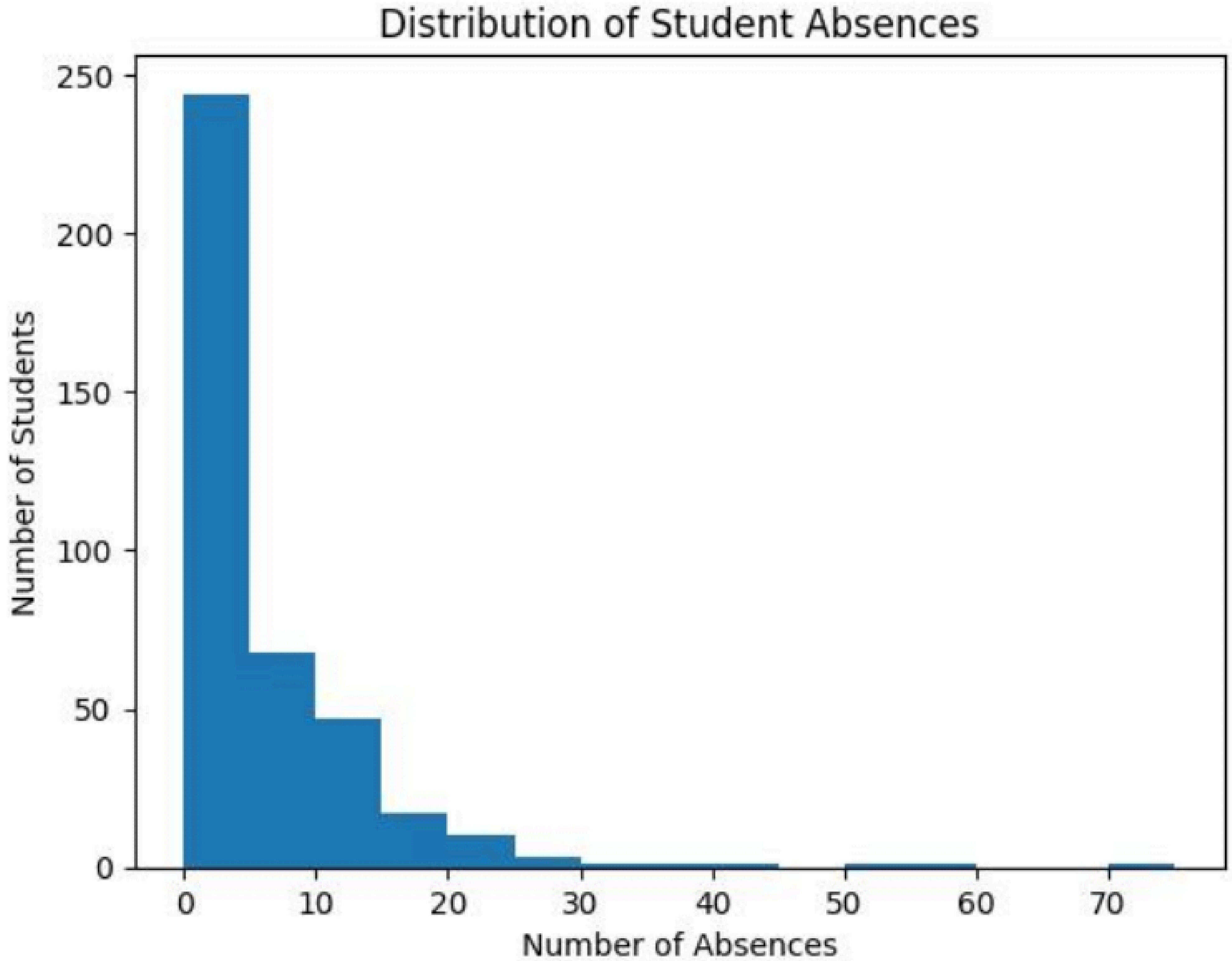


4.3.4 Student Absences

The histogram of student absences showed that most students had relatively few abse nces, while a smaller number recorded high absenteeism.

Higher absenteeism may reduce learning opportunities and negatively affect academic performance.

Figure 4.4: Distribution of Student Absences



4.4 Machine Learning Model Performance

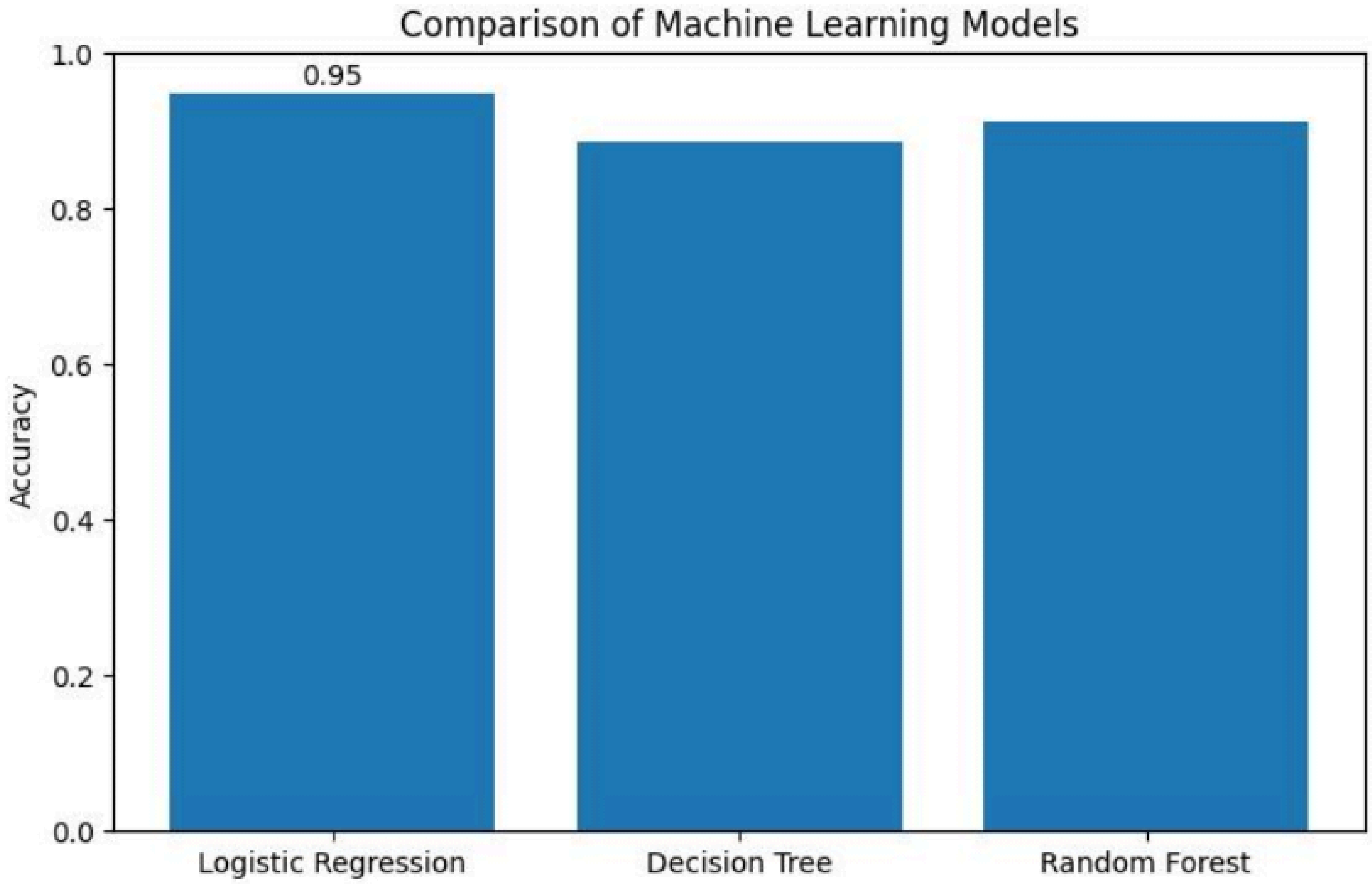
Three supervised machine learning algorithms were trained and evaluated using the same training and testing datasets.

The objective was to determine which algorithm produced the highest prediction accuracy.

Model Comparison

Machine Learning Model	Accuracy
Logistic Regression	94.94%
Random Forest	91.14%
Decision Tree	88.61%

Figure 4.5: Comparison of Machine Learning Model Accuracy



4.5 Discussion of Results

The comparison of the three machine learning algorithms revealed that Logistic Regression achieved the highest prediction accuracy of 94.94%, making it the most suitable model for this dataset.

Although Random Forest also performed well with an accuracy of 91.14%, it was slightly less accurate than Logistic Regression. Decision Tree recorded the lowest accuracy (88.61%), indicating that it was less effective for this particular prediction task.

The high performance of Logistic Regression suggests that the relationship between the selected student features and the target variable was well represented by a linear decision boundary. This made Logistic Regression particularly effective in distinguishing between students who were likely to pass and those who were likely to fail.

The exploratory data analysis also showed that variables such as study time, previous failures, and earlier academic grades (G1 and G2) were strongly associated with final academic performance. These findings are consistent with previous studies in educatio nal data mining, which have shown that historical academic achievement and study be haviour are strong predictors of future examination outcomes.

Although this project used a Mathematics performance dataset due to the unavailabilit y of official WAEC records, the methodology demonstrates how machine learning can b e applied to predict examination outcomes using educational data. With access to offici al WAEC datasets, similar models could support schools and education authorities in id entifying students who may require academic intervention before final examinations.

4.6 Summary

This chapter presented the findings obtained from the exploratory data analysis and th e machine learning models. Logistic Regression emerged as the best-performing algo rithm with an accuracy of 94.94%, outperforming both Random Forest and Decision Tre e. The results demonstrate the potential of machine learning to support educational de cision-making by predicting students' likelihood of passing examinations.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This project developed a machine learning model for predicting students' likelihood of passing examinations under the capstone title "WAEC Pass Likelihood Prediction Using Machine Learning." Due to the unavailability of publicly accessible WAEC candidate dat asets, a publicly available Student Mathematics Performance dataset was used as a ca se study to demonstrate the complete machine learning workflow.

The project followed the standard Data Science lifecycle, including data collection, data understanding, pre-processing, exploratory data analysis (EDA), model development, e valuation, and comparison of machine learning algorithms.

Three classification models—Logistic Regression, Decision Tree, and Random Forest—were trained and evaluated. Their performance was measured using accuracy, confusi on matrices, and classification reports.

The results showed that Logistic Regression achieved the highest accuracy of 94.94%, outperforming Random Forest (91.14%) and Decision Tree (88.61%). This indicates tha t Logistic Regression was the most effective model for predicting student pass likelihood within the selected dataset.

The project also demonstrated that variables such as study time, previous failures, and earlier academic performance (G1 and G2) play important roles in predicting whether a student is likely to pass or fail.

5.2 Conclusion

This study demonstrates that machine learning can effectively predict students' likelihood of passing examinations using educational data. The developed models successfully identified patterns associated with student performance, showing the potential of predictive analytics to support educational decision-making.

Among the three models evaluated, Logistic Regression produced the highest prediction accuracy and was selected as the final model for this project. The findings suggest that machine learning techniques can help educators identify students who may require additional academic support before final examinations.

Although this study used a Mathematics performance dataset because official WAEC data was not publicly available, the methodology remains applicable to actual WAEC datasets when such data becomes available. Therefore, the project achieves its objective of demonstrating how machine learning can be used to predict examination outcomes.

5.3 Recommendations

Based on the findings of this study, the following recommendations are made:

1.Schools should adopt data-driven approaches to identify students who may be at risk of failing examinations at an early stage.

2.Teachers should provide targeted academic support and mentoring for students with previous academic difficulties.

3.Educational institutions should encourage consistent study habits, as increased study time was associated with better academic performance.

4.Schools should maintain comprehensive digital academic records to facilitate future predictive analytics and educational research.

5.Future researchers should validate the proposed approach using official WAEC datasets or larger educational datasets when they become available.

6.Education authorities should explore the integration of machine learning tools into st udent monitoring systems to support evidence-based decision-making.

5.4 Limitations of the Study

The primary limitation of this study is the use of a publicly available Student Mathematics Performance dataset instead of official WAEC examination records. This limitation arose because official WAEC candidate data is not publicly available due to privacy and accessibility constraints.

Additionally, the dataset contains a limited number of student records and may not fully represent the diversity of students across different schools, regions, or examination systems. As a result, the findings should be interpreted as a demonstration of the machine learning methodology rather than a direct prediction of actual WAEC outcomes.

5.5 Future Work

Future research can extend this project by:

1.Using official WAEC datasets where access is available and ethically approved.

2.Incorporating additional factors such as socio-economic status, teacher quality, and school resources.

3.Exploring advanced machine learning and deep learning algorithms, including XGBoost and Artificial Neural Networks.

4.Developing a web-based application that allows teachers or school administrators to input student information and receive pass likelihood predictions in real time.

5.Evaluating the model across multiple subjects to improve its applicability to broader educational settings.

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